A

E-803

# SAMAGRA SHIKSHA KERALA
## SUMMATIVE ASSESSMENT II, 2025 - 2026
# MATHEMATICS

**Class : VIII**

**Time: $1\frac{1}{2}$ Hrs.**

**Total Score : 40**

**Instructions:**

- Use the first 15 minutes to read the questions and think about answers.
- There are 16 questions, split into 4 parts A, B, C, D
- Answer all questions; but in questions of the type A or B you need answer only one of those.
- You can answer the questions in any order, writing the correct question number.
- Answers must be explained whenever necessary.

### Section A

*This section has* ***4 questions of score 1*** *each. Select the correct answer.*

1. Four subtracted from 5 times of a number gives 6. What is the number ?

   A. 1 B. 2 C. 3 D. 0

2. Which of the following gives $-2$ as the answer ?

   A. $7-5$ B. $-7-5$ C. $-5+7$ D. $5-7$

3. Find the missing number in the given table.

| | 20 | 4 |
|---|---|---|
| 20 | 400 | 80 |
| 4 | 80 | ..... |

   A. 160 B. 80 C. 400 D. 16

4. Read the following Statements.

   **Statement I** : The perpendicular bisector of a chord passes through the centre of the circle.

   **Statement II** : Any point at the same distance from the endpoints of a line lies on its perpendicular bisector.

   Choose the correct answer from those given below.

   A. **Statement I** is true, **Statement II** is false.
   B. **Statement I** is false, **Statement II** is true.
   C. Both statements are true, **statement II** is the reason of **statement I.**
   D. Both statements are true, **statement II** is not the reason of **statement I.**

## Section B

*This section has* ***4 questions of score 2*** *each.*

5. A person bought 2 bags and 3 umbrellas for 2300 rupees. The price of 3 umbrellas is 900 rupees.
   i) What is the price of an umbrella ?
   ii) What is the price of a bag ?

6. Draw an angle of $32\frac{1}{2}^0$ .

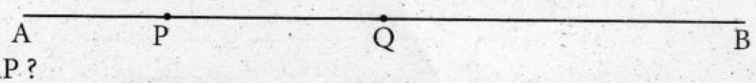
7. Write the decimal form of $\frac{7}{25}$.

8. There are 20 questions in a test. Each correct answer gets 2 marks. For each wrong answer, 1 mark is subtracted. A person answers all questions, but only 5 of them are correct. How much marks he get ?

## Section C

*This section has* ***4 questions of score 3*** *each.*

9. The line AB is divided in the ratio 2:3:5 by the points P and Q.
   QB = 15 centimetres,




   i) a) What is the length of AP ?
      b) What is the length of PQ ?
   ii) If QB = 30 centimetres, what is the length of AP ?

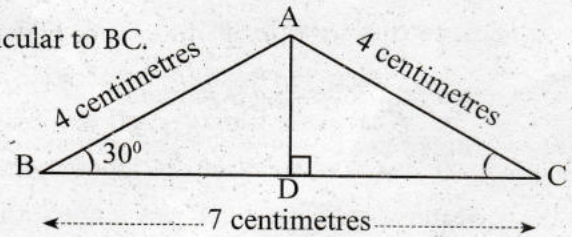
10. A. Prove that the chords of a circle at equal distances from the centre have equal length.

**OR**

B. Prove that the perpendicular from the centre of a circle to a chord bisects the chord.

11. Draw a rhombus of diagonals 7 centimetres, 5.5 centimetres.

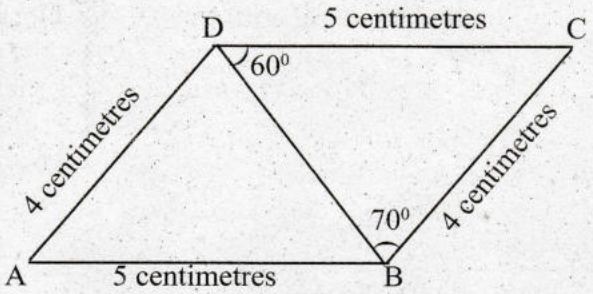
12. A. In the figure, AB = AC = 4 centimetres,
    BC = 7 centimetres, and AD is perpendicular to BC.
    $\angle B = 30^0$




    i) What is $\angle C$ ?
    ii) Write the length of BD.
    iii) What is $\angle BAD$ ?

**OR**

B. In the quadrilateral ABCD, AB = CD = 5 centimetres,
AD = BC = 4 centimetres,
$\angle CDB = 60^0$, $\angle CBD = 70^0$




i) Are the angles of triangle ABD and triangle BCD equal ? Why ?

ii) Write all the angles of triangle ABD.

## Section D

*This section has* ***4 questions of score 4*** *each.*

13. Draw a triangle of two sides 4 centimetres and 6 centimetres; angle between these sides $55^0$. Draw its circumcircle.

14. A. Appu is 5 year older than Ammu. Next year, Appu's age will be twice that of Ammu's age.

| | Age now | Age after one year |
|---|---|---|
| Ammu | $x$ | ..... |
| Appu | $x + 5$ | ..... |

i) Complete the table.

ii) After one year, write the relation between their ages using algebra.

iii) What are their ages now ?

**OR**

B. 60 centimetres long wire is to be bent into a rectangle, with the length 3 centimetres longer than the twice the breadth.

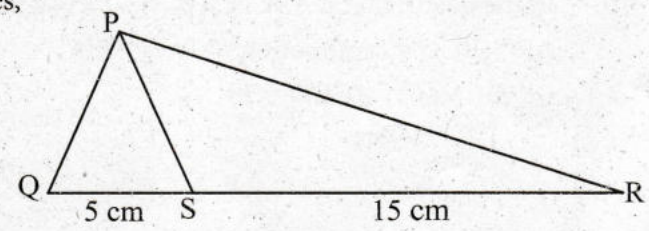
i) What is the sum of length and breadth ?

ii) Form an equation, find the length and breadth.

15. A. Blue and yellow colours are mixed in the ratio 2:3 to make a new colour of 60 litres. To make the ratio 3:4, should more blue or yellow to be added ? How many litres ?

**OR**

B. In the figure QS = 5 centimetres,
SR = 15 centimetres.




i) What is QS:SR ?

ii) Write the ratio of areas of triangle PQS and triangle PSR.

iii) The area of triangle PQS is 10 square centimetres.
What is the area of triangle PQR ?

16. A regular polygon has 10 sides, then

(i) a) What is the sum of its outer angles ?

b) Find the outer angle and inner angle at one vertex.

(ii) Name the polygon in which inner angles and outer angles are equal ?